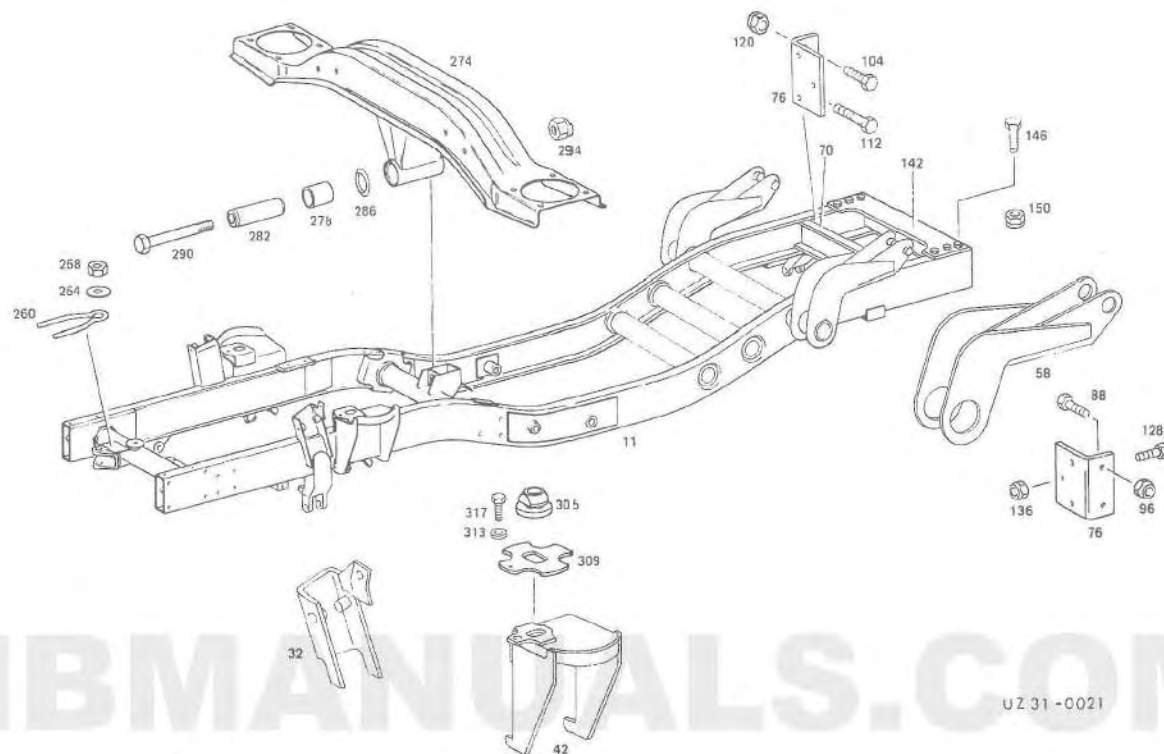


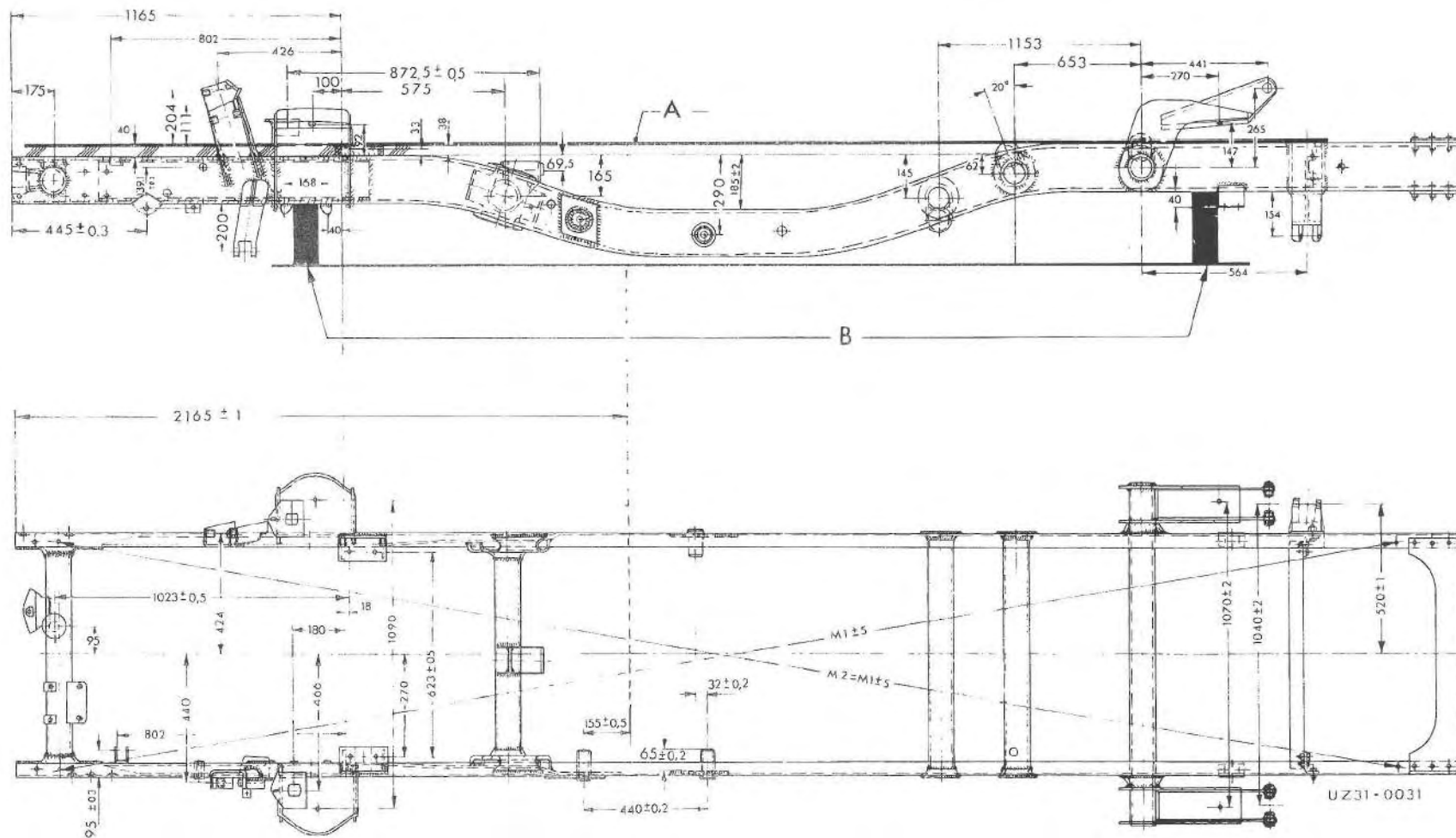
Exploded view



Frame

11	Frame	150	Nut
32	Bracket	260	Retainer
42	Spring bracket	264	Washer
58	Spring bracket	268	Nut
70	Crossmember	274	Body support
76	Bracket	282	Bushing
88	Bolt	286	O-ring
96	Nut	290	Bolt
104	Bolt	294	Nut
112	Bolt	305	Rubber mount
128	Bolt	309	Packing
136	Nut	313	Washer
142	Crossmember	317	Bolt
146	Bolt		

General view



Frame 435.117 (standard)

A Straight edge (beam test)

B Supporting points

Contents

Chapter	Page
1 General	
Installation survey	1.1/1
2 Checking frame alignment	2.1/1

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Installation survey

Chassis Model	Sales designation	Frame Manufacturer	Installation Standard	SA
435.110	U 1700	Daimler-Benz	X	
.111	U 1700 L			
.113	U 1700 L/38			35 999

Contents

Chapter	Page
1 General	
Installation survey	1.1/1
Technical data	1.1/1

Survey

Version

Springs 435.115/117

32.6

Springs 435.110/111/113/160/170

32.7

Shock absorbers 435.115/117 and 435.110/111/113/160

32.11

Torsion bar stabilizer

32.21**MBMANUALS.COM**

Contents

Chapter	Page
1 General	
Installation survey	1.1/1
Technical data	1.1/1
Exploded view	1.2/1

Installation Survey

Chassis Model	Sales designation	Spring Front	Auxiliary	Rear	Auxiliary	Installation
435.115	U 1300 L	435 321 02 04	—	435 324 02 04 ¹⁾	435 324 03 07 ¹⁾	Series
.117	U 1300 L/37			435 324 05 04 ²⁾		
435.115	U 1300 L	435 321 03 04	—	435 324 04 04	—	SA 35 252

1) Valid to chassis end No. 088 157

2) Valid from chassis end No. 088 158

Technical Data

Front spring 435 321 02 04

Spring gauge $\varnothing$	d	27,5 + 0,2 mm
Extended spring	L _o	376 ± 12 mm
Max. block length	L _{B L}	161 mm
Direction of winding		right-hand
Effective turns	I _t	4 $\frac{1}{8}$
Total turns	I _g	6 $\frac{3}{8}$

Size class code

	L _N in 15 700 N	Yellow line
Class 1	242 $\begin{smallmatrix} -1 \\ -8 \end{smallmatrix}$	I
Class 2	242 $\begin{smallmatrix} +3 \\ -1 \end{smallmatrix}$	II
Class 3	242 $\begin{smallmatrix} +8 \\ +3 \end{smallmatrix}$	III

32.6 General

Rear Spring (with auxiliary spring) 435 324 02 04 ¹⁾

Spring gauge $\varnothing$	d	27 - 0,2 mm
Extended spring	L _o	437 ± 12 mm
Max. block length	L _{BL}	196 mm
Direction of winding		right-hand
Effective turns	i _f	6
Total turns	i _g	7,5

1) Valid from chassis No. 435 115- 088 157

Size class code ²⁾

	L _N in 14 710 N	Yellow line
Class 1	288 $\begin{smallmatrix} -1 \\ -8 \end{smallmatrix}$	I
Class 2	288 $\begin{smallmatrix} +3 \\ -1 \end{smallmatrix}$	II
Class 3	288 $\begin{smallmatrix} +8 \\ +3 \end{smallmatrix}$	III

2) Valid to chassis end No. 435 115 - 001 960

Size class code ³⁾

	L _N in 7850 N	Green line
Class 1	356 $\begin{smallmatrix} -6 \\ -10 \end{smallmatrix}$	I
Class 2	356 $\begin{smallmatrix} -2 \\ -6 \end{smallmatrix}$	II
Class 3	356 $\begin{smallmatrix} +2 \\ -2 \end{smallmatrix}$	III

3) Valid from chassis end No. 435 115 - 001 961

Rear Spring (without auxiliary spring) 435 324 05 04 ²⁾

Spring gauge $\varnothing$	d	30,5 mm
Extended spring	L _o	482 ± 10 mm
max. block length	L _{BL}	235 mm
Direction of winding		right-hand
Effective turns	i _f	7,125
Total turns	i _g	8,625

2) Valid from chassis No. 435 115- 088 158

Size class code

	L _N in 16 700 N	Yellow line
Class 1	338 ⁻⁶ -10	I
Class 2	338 ⁻² -6	II
Class 3	338 ⁺² -2	III

Auxiliary spring rear 435 324 03 07 ¹⁾

Spring gauge $\varnothing$	d	18,5 mm
Extended spring	L _o	275 ± 10 mm
Max. block length	L _{BL}	133 mm
Direction of winding		left-hand
Effective turns	i _f	7
Total turns	i _g	8,5

1) Valid to chassis No. 435 115- 088 157

32.6 General

SA 35 252

Front Spring, reinforced 435 321 03 04

Spring gauge $\varnothing$	d	$25,4 \pm 0,1$ mm
Extended spring	L_o	452 ± 12 mm
Max. block length	L_{BL}	165 mm
Direction of winding		right-hand
Effective turns	i_f	5 $\frac{3}{8}$
Total turns	i_g	6 $\frac{1}{8}$

Size class code

	L_N in 14 710 N	Yellow line
Class 1	260 $\begin{smallmatrix} -1 \\ -8 \end{smallmatrix}$	I
Class 2	260 $\begin{smallmatrix} +3 \\ -1 \end{smallmatrix}$	II
Class 3	260 $\begin{smallmatrix} +8 \\ +3 \end{smallmatrix}$	III

Rear Spring, reinforced 435 324 04 04

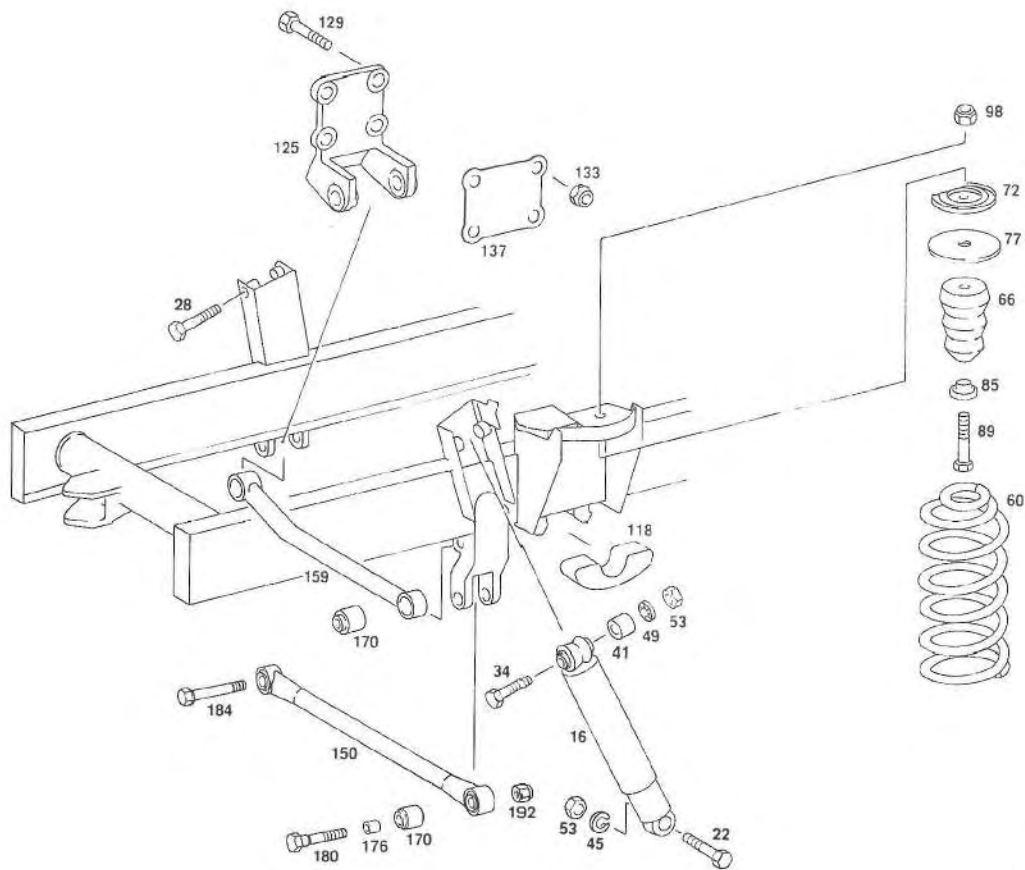
Spring gauge $\varnothing$	d	27,2 $\pm$ 0,1 mm
Extended spring	L _o	552 $\begin{smallmatrix} + 6 \\ - 12 \end{smallmatrix}$ mm
Max. block length	L _{BL}	232 mm
Direction of winding		right-hand
Effective turns	i _f	7 $\frac{5}{8}$
Total turns	i _g	9 $\frac{5}{8}$

Size class code

	L _N in 15,690 N	Yellow line
Class 1	340 $\begin{smallmatrix} - 6 \\ - 10 \end{smallmatrix}$	I
Class 2	340 $\begin{smallmatrix} - 2 \\ - 6 \end{smallmatrix}$	II
Class 3	340 $\begin{smallmatrix} + 2 \\ - 2 \end{smallmatrix}$	III

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Exploded View

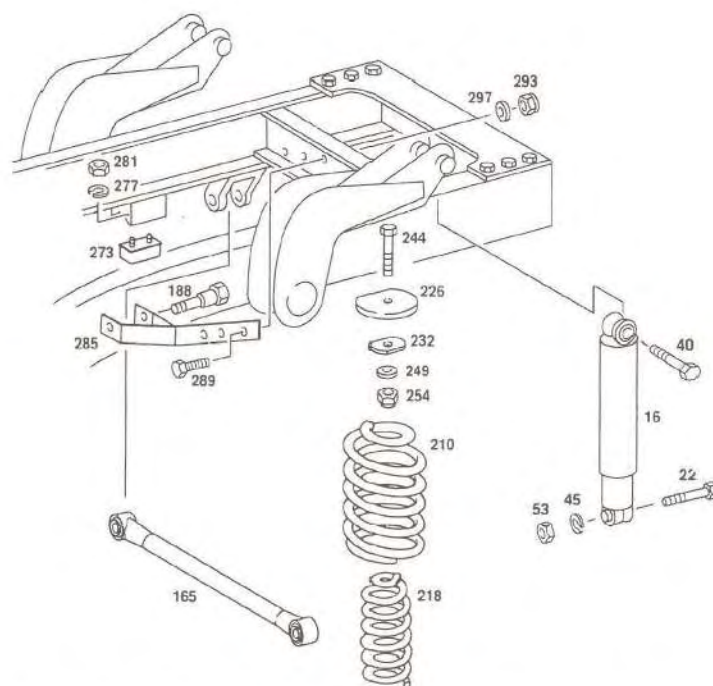


UZ32-0015

Springs and Shock Absorber Front

16	Shock absorber	118	Rubber buffer
22	Bolt	125	Bearing bracket
28	Bolt	129	Bolt
34	Bolt	133	Nut
41	Bearing	137	Gasket
45	Lock washer	150	Transverse link
49	Lock washer	159	Guide rod
53	Nut	170	Bearings
60	Front spring	176	Bearing bushing
72	Plate	180	Bolt
89	Bolt	184	Bolt
98	Nut	192	Nut

Exploded View



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UZ32-0016

Springs and Shock Absorber Rear

16	Shock absorber	244	Bolt
22	Bolt	249	Washer
40	Bolt	254	Nut
45	Lock washer	273	Rubber buffer
53	Nut	277	Lock washer
165	Transverse link	281	Nut
188	Bolt	285	Strut
210	Rear spring	289	Bolt
218 ¹⁾	Auxiliary spring	284	Nut
226	Plate	293	Washer
232	Plate	297	Bolt

1) Valid to chassis No. 435 115 – 088 157

Contents

Chapter	Page
1 General	
Installation Survey	1.1/1
Technical Data	1.1/1
Exploded View	1.2/1

Installation survey

Chassis Model	Sales designation	Spring Front	Auxiliary	Spring Rear	Auxiliary	Installation
435.110	U 1700	425 321 07 04	000 321 00 06	425 324 03 04 ¹⁾	425 324 02 06 ¹⁾	Series
.111	U 1700 L			425 324 04 04 ²⁾	425 324 00 06 ²⁾	
.113	U 1700 L/38			435 324 06 04 ¹⁾	435 324 00 06 ¹⁾	
.110	U 1700	425 321 08 04	000 321 00 06	425 324 03 04 ¹⁾	425 324 02 06 ¹⁾	SA 35 301
.111	U 1700 L			425 324 04 04 ²⁾	425 324 03 06 ²⁾	
.160	Kit TM 170	425 321 08 04	425 320 00 09 ³⁾	435 324 06 04	435 324 00 06	Series
.170	Kit Condor	425 324 04 04	425 324 03 06	425 324 04 04	425 324 03 06	

1) Valid to chassis No. 435 111 – 001 086

2) Valid from chassis No. 435 111 – 001 087

3) Seal spring

Technical data

Front spring 425 321 07 04

Spring gauge Ø	d	26,5 mm
Extended spring	L _o	405 ± 12 mm
Max. block length	L _{B L}	170 mm
Direction of winding		right-hand

Size class code

	L _N in 17 600 N	Yellow line
Class 1	257 $\begin{smallmatrix} -1 \\ -8 \end{smallmatrix}$	I
Class 2	257 $\begin{smallmatrix} +3 \\ -1 \end{smallmatrix}$	II
Class 3	257 $\begin{smallmatrix} +8 \\ +3 \end{smallmatrix}$	III

32.7 General

Rear spring 425 324 03 04

Spring gauge $\varnothing$	d	27 mm
Extended spring	L_o	485 ± 12 mm
Max. block length	L_{BL}	208 mm
Direction of winding		right-hand
Effective turns	i_f	6,75
Total turns	i_g	8,25

Size class code

	L_N in 18 640 N	Yellow line
Class 1	295 $\begin{smallmatrix} -1 \\ -8 \end{smallmatrix}$	I
Class 2	295 $\begin{smallmatrix} +3 \\ -1 \end{smallmatrix}$	II
Class 3	295 $\begin{smallmatrix} +8 \\ +3 \end{smallmatrix}$	III

Rear spring 425 324 02 06

Spring gauge $\varnothing$	d	18,5 mm
Extended spring	L_o	310 ± 8 mm
Max. block length	L_{BL}	165 mm
Direction of winding		right-hand
Effective turns	i_f	8
Total turns	i_g	9,5

Rear spring 435 324 06 04

Spring gauge $\varnothing$	d	29,5 mm
Extended spring	L_o	502 ± 12 mm
Max. block length	L_{BL}	216 mm
Direction of winding		right-hand

Size class code

	L_N in 22 600 N	Yellow line
Class 1	$312 \begin{smallmatrix} -1 \\ -8 \end{smallmatrix}$	I
Class 2	$312 \begin{smallmatrix} +3 \\ -1 \end{smallmatrix}$	II
Class 3	$312 \begin{smallmatrix} +8 \\ -8 \end{smallmatrix}$	III

Rear spring 435 324 00 06

Spring gauge $\varnothing$	d	20,5 mm
Extended spring	L_o	340 ± 10 mm
Max. block length	L_{BL}	170 mm
Direction of winding		right-hand

Size class code

	L_N in 8300 N	Yellow line
Class 1	$256 \begin{smallmatrix} -1 \\ -8 \end{smallmatrix}$	I
Class 2	$256 \begin{smallmatrix} +3 \\ -1 \end{smallmatrix}$	II
Class 3	$256 \begin{smallmatrix} +8 \\ +3 \end{smallmatrix}$	III

32.7 General

SA 35 301

Rear spring 425 324 04 04

Spring gauge $\varnothing$	d	28,8 mm
Extended spring	L _o	488 ± 12 mm
Max. block length	L _{B L}	211 mm
Direction of winding		right-hand

Size class code

	L _N in 21 582 N	Yellow line
Class 1	300 $\begin{smallmatrix} -1 \\ -8 \end{smallmatrix}$	I
Class 2	300 $\begin{smallmatrix} +3 \\ -1 \end{smallmatrix}$	II
Class 3	300 $\begin{smallmatrix} +8 \\ +3 \end{smallmatrix}$	III

Front spring 425 321 08 04

Spring gauge $\varnothing$	d	28,3 mm
Extended spring	L _o	426 ± 12 mm
Max. block length	L _{B L}	180 mm
Direction of winding		right-hand

Size class code

	L _N in 20 600 N	Yellow line
Class 1	265 $\begin{smallmatrix} -1 \\ -8 \end{smallmatrix}$	I
Class 2	265 $\begin{smallmatrix} +3 \\ -1 \end{smallmatrix}$	II
Class 3	265 $\begin{smallmatrix} +8 \\ +3 \end{smallmatrix}$	III

Auxiliary spring 425 324 03 06

Spring gauge $\varnothing$	d	19,5 mm
Extended spring	L_o	328 ± 8 mm
Max. block length	L_{BL}	168 mm
Direction of winding		right-hand

Size class code

	L_N in 9 810 N	Yellow line
Class 1	239 $\begin{smallmatrix} -1 \\ -5 \end{smallmatrix}$	I
Class 2	239 $\begin{smallmatrix} +2 \\ -1 \end{smallmatrix}$	II
Class 3	239 $\begin{smallmatrix} +5 \\ +2 \end{smallmatrix}$	III

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425 324 03 06

Spring wire ϕ	d	19.5 mm
Extended spring length	L_0	328 ± 8 mm
Max. block length	L_{BL}	168 mm
Direction of turn		right-hand
Effective turns	i_f	8.5
Total turns	i_g	10
Linear turns	i_l	4.5
Progressive turns	i_p	4
	L_1	300 ± 7 mm at 1960 N
Spring test L_N (mm) at F_N (N)	L_N	239 ± 5 mm at 9810 N
	L_2	196 ± 7 mm at 16 680 N

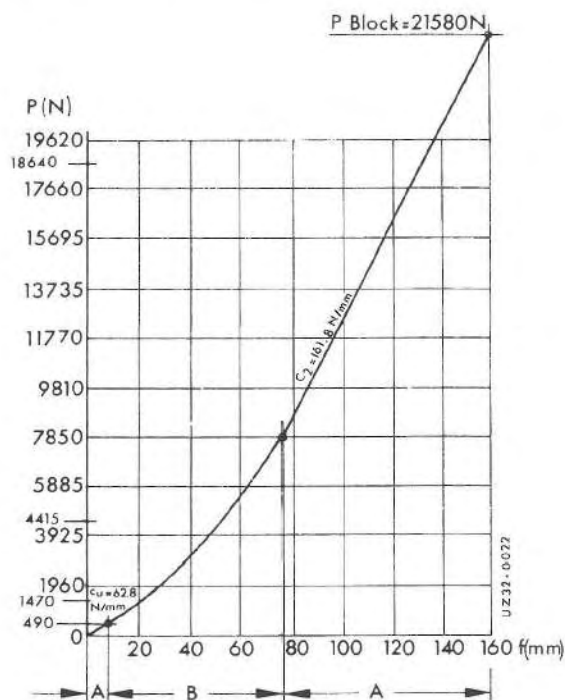
Size grade marking

	L_N at 9810 N	yellow line
Grade 1	$239 \begin{smallmatrix} -1 \\ -5 \end{smallmatrix}$	I
Grade 2	$239 \begin{smallmatrix} +2 \\ -1 \end{smallmatrix}$	II
Grade 3	$239 \begin{smallmatrix} +5 \\ +2 \end{smallmatrix}$	III

Test specifications

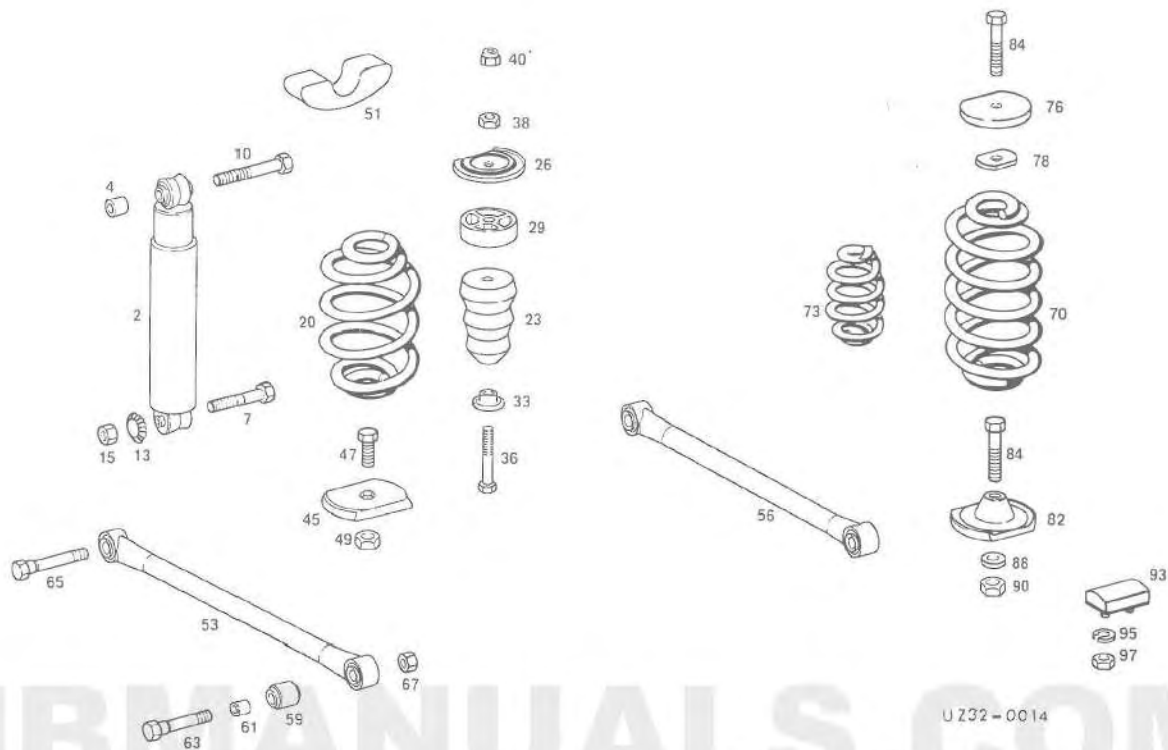
1. Compress spring to theoretical block length L_{BL} .
2. Check test lengths L_1 , L_N and L_2 .
3. Release spring and check length L_0 .

Spring characteristic



A = Linear
 B = Progressive
 C_u/C_2 = Spring constant

Exploded View



Springs and Shock Absorber

2	Shock absorber front and rear	53	Transverse link, front
4	Bushing	56	Transverse link, rear
7	Bolt	59	Bearings
10	Bolt	61	Bearing bushing
13	Lock washer	63	Bolt
15	Nut	65	Bolt
20	Front spring	67	Nut
23	Auxiliary spring front	70	Rear spring
26	Plate	73	Auxiliary spring, rear
29	Spacer	76	Plate
33	Spring sleeve	78	Plate
36	Bolt	82	Spring seat
38	Nut	84	Bolt
40	Nut	88	Washer
45	Spring seat	90	Nut
47	Bolt	93	Rubber buffer
49	Nut	95	Lock washer
51	Rubber buffer	97	Nut

Contents

Chapter	Page
1 General	
Installation Survey	1.1/1
Technical Data	1.1/1

Installation survey

Chassis		Shock absorber			
Model	Sales designation	Manufacturer	front/rear		Installation
435.115	U 1300 L	Fichtel & Sachs		004 323 26 00	Series
.117	U 1300 L/37	Boge	optional	004 322 27 00	
.110	U 1700	Fichtel & Sachs Boge		003 323 15 00	
.111	U 1700 L			002 323 93 00	
.113	U 1700 L/38		optional	003 323 70 00 003 323 93 00	
.115	U 1300 L	Boge		003 323 35 00 ¹⁾	SA 35 633 ³⁾
.117	U 1300 L/37		optional	003 323 36 00 ²⁾ 003 323 78 00	
.113	U 1700 L/38	Fichtel & Sachs		003 323 86 00 Series	
.160	Kit TM 170				

1) To chassis No. 435 115 - 000 043

2) From chassis No. 435 115 - 000 044

3) Shock absorber reinforced

Technical Data

Shock absorber 004 323 27 00

Cylinder Ø	mm	64
Piston Ø	mm	58
Bore Ø	mm	20 H 9
Minimum length	mm	437 ⁺² ₋₅
Maximum length	mm	707 ⁺⁵ ₋₂
Hydraulic stop effective at	mm	690

Shock absorber 004 323 26 00

Cylinder Ø	mm	62
Piston Ø	mm	55
Bore Ø	mm	20 H 9
Minimum length	mm	440 ± 3
Maximum length	mm	710 ± 3
Hydraulic stop effective at	mm	704

Shock absorber 003 323 15 00

Cylinder Ø	mm	75
Piston Ø	mm	65
Bore Ø	mm	20 H 9
Minimum length	mm	468 ± 3
Maximum length	mm	753 ± 3
Hydraulic stop effective at	mm	744

32.11 General

Shock absorber 002 323 93 00

Cylinder Ø	mm	83
Piston Ø	mm	72
Bore Ø	mm	20 H 9
Minimum length	mm	458^{+2}_{-5}
Maximum length	mm	758^{+5}_{-3}
Hydraulic stop effective at	mm	735

Shock absorber 003 323 35 00 ¹⁾

Cylinder Ø	mm	83
Piston Ø	mm	72
Bore Ø	mm	20 H 9
Minimum length	mm	458^{+2}_{-5}
Maximum length	mm	748^{+5}_{-2}
Hydraulic stop effective at	mm	735

1) Valid to chassis No. 435 115 - 000 043

Shock absorber, reinforced 003 323 36 00 ²⁾

Cylinder Ø	mm	83
Piston Ø	mm	72
Bore Ø	mm	20 H 9
Minimum length	mm	442^{+2}_{-5}
Maximum length	mm	707^{+5}_{-2}
Hydraulic stop effective at	mm	694

2) Valid from chassis No. 435 115 - 000 044

Shock absorber, reinforced 003 323 78 00

Cylinder Ø	mm	75
Piston Ø	mm	65
Bore Ø	mm	20 H 9
Minimum length	mm	445 ± 3
Maximum length	mm	710 ± 3
Hydraulic stop effective at	mm	702

Shock absorber, reinforced 003 323 86 00

Cylinder Ø	mm	75
Piston Ø	mm	65
Bore Ø	mm	20 H 9
Minimum length	mm	471 ± 3
Maximum length	mm	753 ± 3
Hydraulic stop effective at	mm	744

Shock absorber, reinforced 003 323 70 00

Cylinder Ø	mm	75
Piston Ø	mm	65
Bore Ø	mm	20 H 9
Minimum length	mm	471 ± 3
Maximum length	mm	753 ± 3
Hydraulic stop effective at	mm	744

Shock absorber, reinforced 003 323 93 00

Cylinder Ø	mm	82
Piston Ø	mm	72
Bore Ø	mm	20 H 9
Minimum length	mm	462 ⁺² ₋₅
Maximum length	mm	747 ⁺⁵ ₋₂
Hydraulic stop effective at	mm	734

Contents

Chapter	Page
1 General	
Installation Survey	1.1/1
Tightening Torques	1.1/1
Necessary Material	1.1/1
Exploded View	1.2/1
2 Installation of Torsion Bar - Stabilizer	2.1/1
3 Renewing plastic mount	3.1/1
Removal of Plastic Mount	3.1/1
Installation of Plastic Mount	3.1/2

Installation Survey

Chassis Model	Sales Designation	Torsion bar diameter in mm				Installation
		front Ø 36	reinforced Ø 40	rear Ø 36	reinforced Ø 40	
435.110 .111	U 1700 U 1700 L	425 323 03 65	435 323 02 65	387 323 01 65	435 323 00 65	SA 35 772
.113	U 1700 L/38			—		
.115 .117	U 1300 L U 1300 L/37		—	387 323 01 65		
.160	Kit TM 160		435 323 02 65	—	435 323 00 65	Series

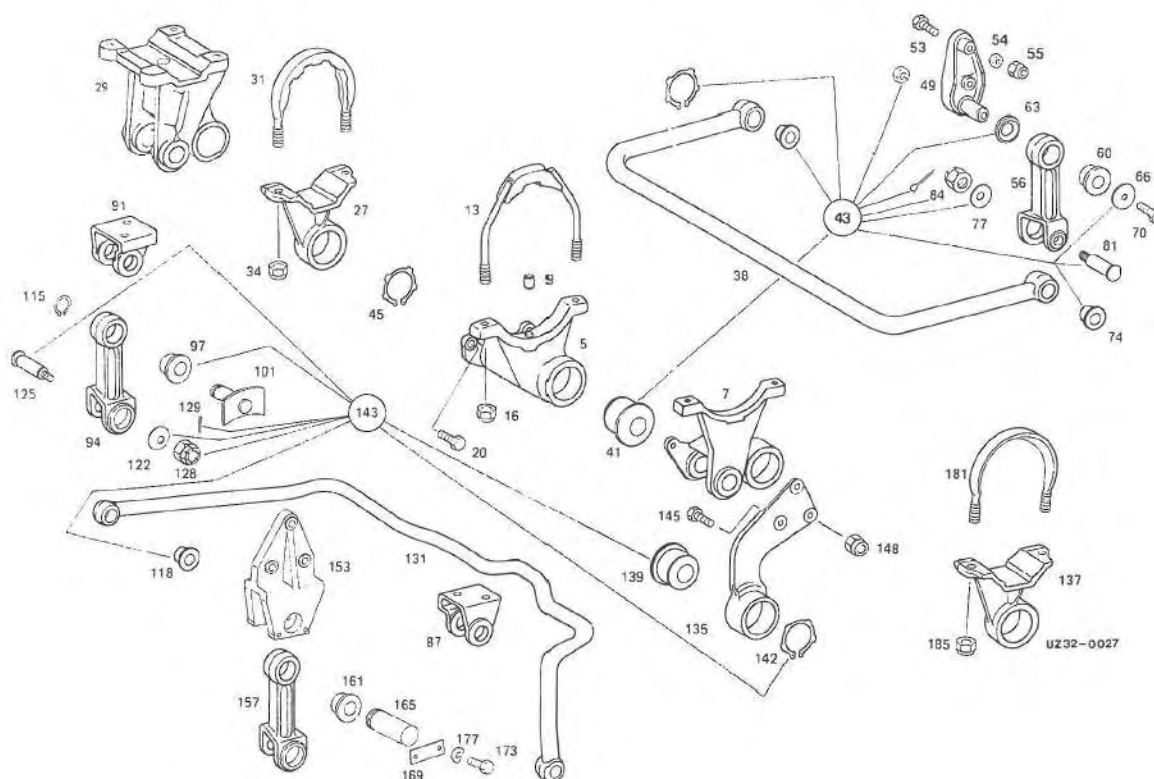
Tightening torques

Designation	Stabilizer		Thread strength	Nm
	front	rear		
Bearing bracket to frame	X		M 12 x 1,5 – 10,9	140
Bearing bracket to axle tube	X		M 18 x 1,5 – 10,9	350
Bearing bracket to frame		X	M 14 x 1,5 – 10,9	180
Bracket to axle housing		X	M 14 x 1,5 – 12,9	200
Stirrup bolt to axle tube left		X	M 14 x 1,5	170
Stirrup bolt to axle tube right		X	M 16 x 1,5	170

Necessary Materials

Cons. No.	Designation	Part number
1	Molycote - liquid grease	commercially available

Exploded View



Torsion bar - stabilizer for front and rear axle

5	Bracket, rear left	91	Bearing bracket, front right
7	Bracket, rear left	94	Connecting rod, front
9	Pin	97	Bushing, front
13	Clamp	101	Pin
16	Nut	118	Bushing
20	Screw	122	Belleville spring washer
27	Bearing bracket rear right	125	Pin
29	Bearing bracket rear right	128	Nut
31	Clamp	129	Cotter pin
34	Nut	131	Torsion bar, front
38	Torsion bar	135	Bearing bracket, front
41	Rubber mount	137	Bearing bracket, rear
43	Repair kit, rear	139	Rubber mount
45	Retaining clip	142	Retaining ring
49	Bearing bracket, rear	143	Repair kit front
53	Screw	145	Screw
54	Nut	148	Nut
55	Nut	153	Bearing bracket, rear
56	Connecting rod, rear	157	Connection rod, front
60	Bushing	161	Bushing
70	Screw	165	Pin
74	Bushing	169	Retainer
77	Belleville spring washer	173	Screw
81	Pin	177	Spring washer
84	Nut	181	Clamp
87	Bearing bracket front left	185	Nut

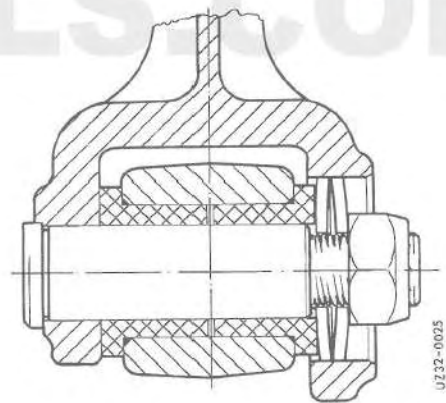
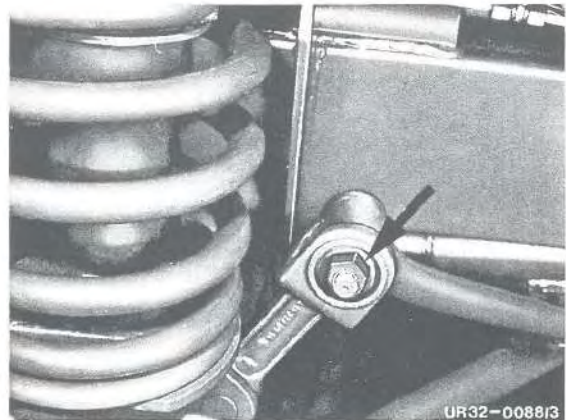
Installation Notes:

- Only use repair kits for repairing bearing mounts
- Grease all plain bearings with molycote liquid grease No. 1.

- Prior to assembly, coat plain bearing and bearing bushing as well as belleville spring washers with molycote liquid grease.

- Ensure the belleville spring washers are correctly installed.

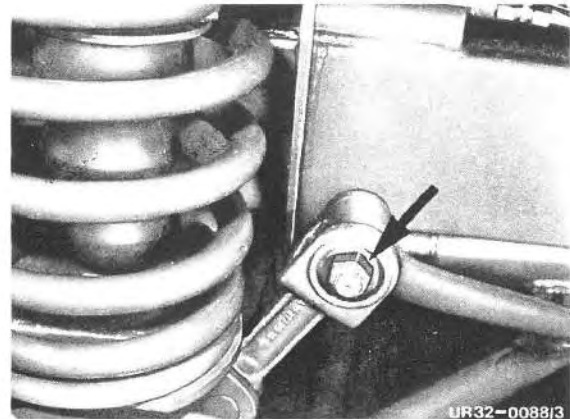
- Slightly tighten the securing nut.
The plastic bearing mounts must not be subjected to pressure.



Removal

- Unscrew nut.

Installation Note: Screw in nut ensuring that no pressure is applied to the plastic bearings.



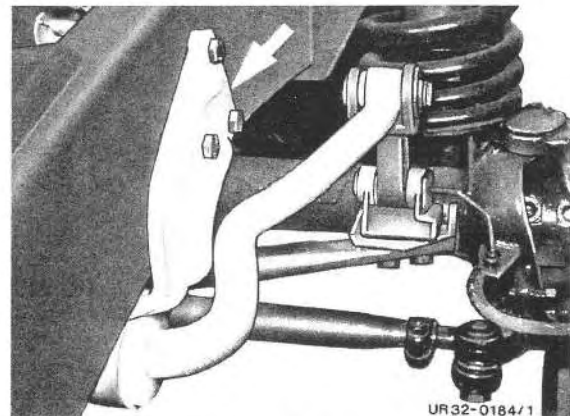
- Remove spring washers and bearing bush.
- Drive out pin.

Installation Note: Coat bearing bush with Molykote No. 1.



- Unscrew bracket on axle housing
Tightening torque!

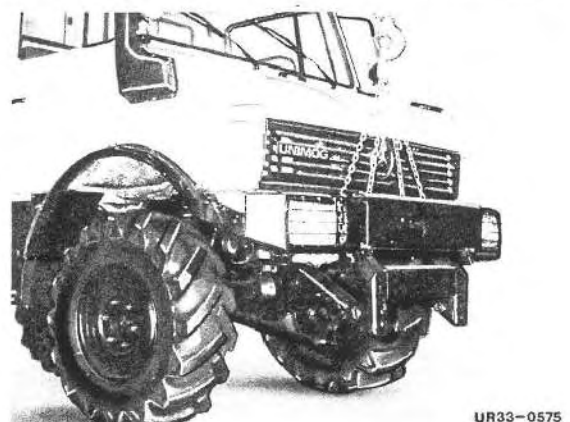
Installation Note: Fit bolts with sealing compound No.2.



- Unscrew nuts
Tightening torque
- Remove stabilizer

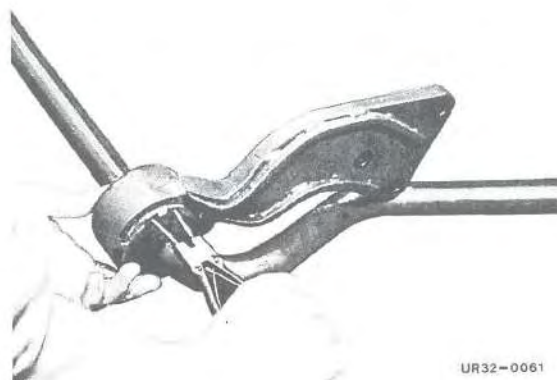
Installation

- Install in the reverse order of removal.

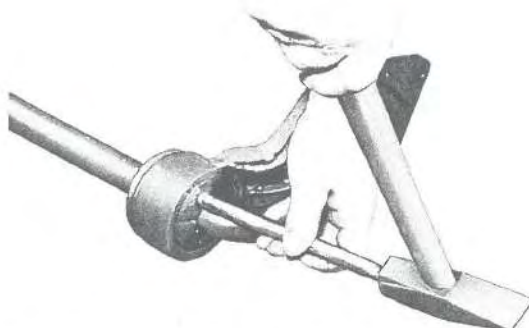


Removal

- Remove circlip.



- Knock out bearing bushing.



- Press apart bearing bushing with an assembly lever, or saw off.



32.21 Removal and Installation of Plastic Bearing Mount

SA 35 772

Installation

Note: Only use repair kits for repair work.

– Grease bearing point of torsion bar with molycote liquid grease No. 1.

– Press on bearing using forge tongs and install on torsion bar.

– Press bearing together with screw clamp and insert in bearing bracket.



– Install circlip.

Note: Ensure circlip is correctly seated.

